Ahsan Ahmed

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Education

Bangladesh University of Engineering and Technology (BUET) **Dhaka, Bangladesh**

Bachelor of Science in Naval Architecture and Marine Engineering *Jan. 2022 – Jun. 2026*

- **CGPA: 3.57 / 4.00** (3.89 in last semester)
- **Class Rank:** 8 / 55
- Major: Hydrodynamics and Structure
- **Relevant Courses:** Fluid Mechanics · Marine Hydrodynamics · Computational Fluid Dynamics (CFD) · Advanced Ship Structure · Ship Vibration · Finite Element Method (FEM) for Ship Structure · Resistance and Propulsion of Ships · Heat Transfer · Power and Propulsion Systems

Research Interest

- **Computational Fluid Dynamics (CFD)**
- **Fluid–Structure Interaction**
- **Ship Hydrodynamics**
- **Smoothed Particle Hydrodynamics (SPH)**
- **Free-Surface and Multiphase Flows**
- **Fluid Mechanics**

Research Experience

- **Ongoing Research** *Sep. 2026 – Present*

  Project title: *Prediction of Added Resistance and Heave–Pitch Motions of the KCS Hull in Regular Head Waves Using Smoothed Particle Hydrodynamics*

  Under the supervision of: Dr. Md. Mashiur Rahaman, Dept. of NAME, BUET

  - Extending the validated calm-water SPH model to regular head waves to predict added resistance and heave–pitch motions of the KCS hull in DualSPHysics.
  - Validating predictions against published towing tank data and comparing free-surface flow features with the calm-water condition.

- **Undergraduate Thesis** *Apr. 2025 – Jun. 2026*

  Thesis title: *Prediction of Calm Water Resistance of a Ship Using Smoothed Particle Hydrodynamics (SPH)*

  Under the supervision of: Dr. Md. Mashiur Rahaman, Dept. of NAME, BUET

  - Applied the mesh-free SPH method to predict calm-water resistance of the KCS hull using the GPU-accelerated DualSPHysics solver.
  - Developed a numerical towing tank and conducted a convergence study by varying particle resolutions (8 mm to 11 mm), identified 9 mm as the optimal spacing.
  - Processed simulation output via Python and ParaView, validating calm-water resistance against experimental towing tank data across Fr = 0.152–0.282 with **errors under 6.5%**.

- **13th Annual International Student Design Competition for Safe and Affordable Ferry organized by Worldwide Ferry Safety Association – 3rd Place (Global)** *Sep. 2025 – Mar. 2026*

  Project title: *Designing a RoPax ferry for the Niger River in Nigeria that can accommodate 200 passengers and their personal and market cargo from Lokoja to the trading city of Onitsha*

  Under the supervision of: Dr. Zobair Ibn Awal, Dept. of NAME, BUET

  - Performed structural scantling calculations using Bureau Veritas (BV) rules for an Aluminium-5086 longitudinally framed catamaran, designed the midship section, prepared a detailed lightship weight estimate, and verified longitudinal strength under normal and grounding conditions.
  - Conducted intact stability, probabilistic damage stability, floodable length, and seakeeping analyses in MAXSURF to verify compliance with IMO/SOLAS criteria.

Manuscripts in Preparation

- S. P. Gupto, **A. Ahmed**, S. S. S. Shouvik, and M. M. Rahaman, "Comparative Assessment of Accuracy and Computational Efficiency of Mesh-Based and Mesh-Free CFD Methods for Ship Resistance Prediction," *in preparation*, 2026.

Conference Presentations

- M. K. M. Zidan, M. A. Rahaman, M. K. Islam, S. P. Gupto, **A. Ahmed**, S. Tubash, and A. Mondal, team presentation on the design of the *Niger Drift* passenger ferry, *Worldwide Ferry Safety Association (WFSA) Ferry Safety and Technology Conference*, May 2026.

Academic Projects

- **Junior Year Project** *Jan. 2024 – Feb. 2025*

  Project title: *Design of an Inland Passenger Ferry with the Capacity of 50 Cars and 700 Passengers for the Route Dhaka–Hatiya–Dhaka*

  Under the supervision of: Dr. Goutam Kumar Saha, Dept. of NAME, BUET

  - Performed hydrostatic, stability, resistance, and powering analyses; carried out scantling calculations following ABS rules with 2D AutoCAD drawings; developed a 3D CAD model (Rhinoceros) and fabricated a scaled prototype via 3D printing.
  - Incorporated a mechanically driven Z-drive azimuthal propulsion system in place of a conventional shaft–rudder arrangement for improved maneuverability and shallow-water operation on the narrow, congested route.

Professional Experience

Intern *Jun. 2024 – Jul. 2024*

Dockyard & Engineering Works Ltd., Bangladesh Navy *Narayanganj, Bangladesh*

- Observed CNC machining, welding processes, marine propulsion systems, shipboard air conditioning, thermal management of engine rooms, and core shipbuilding operations (launching, berthing, piping, painting).
- Developed understanding of shipyard management practices, including tendering processes and multidisciplinary coordination.

Technical Skills

Simulation Software	DualSPHysics, ANSYS Fluent, ABAQUS
Programming Languages	Python, C, C++, Fortran, MATLAB
CAD & 3D Modelling	Rhinoceros 3D, Orca3D, AutoCAD
Ship Design Software	MAXSURF, DELFTship
Tools	ParaView, LaTeX, Zotero, Microsoft Office (Word, Excel, PowerPoint)

Awards and Achievements

- **3rd Place (Global) and recipient of $1,000 in prize money**, 13th Annual International Student Design Competition for Safe and Affordable Ferry, Worldwide Ferry Safety Association (WFSA) ***Mar. 2026***
- **2nd Place and recipient of BDT 2,000 in prize money**, NAME Olympiad (Senior Category), NAME FEST, BUET ***2026***
- **University Merit Scholarship, BUET**, awarded for academic excellence ***2022, 2024, 2025***
- **Education Board Scholarship** in the Higher Secondary Certificate (HSC) Examination ***2020***
- **Education Board Scholarship** in the Secondary School Certificate (SSC) Examination ***2018***

Certifications & Training

- **Rhino 8** (40.5 hours), Udemy ***Sep. 2026***
- **Fire Fighting, Fire Prevention, Rescue and First Aid Training**, Bangladesh Fire Service & Civil Defence Directorate, held at Dept. of NAME, BUET ***Jan. 2026***

Extracurricular Activities

- **Volunteer**, NAME FEST, Dept. of NAME, BUET ***2023, 2026***
- **Member**, Badhan (BUET Unit), voluntary blood donation organization
- **Member**, BUET Automobile Club, BUET Career Club, BUET Photographic Society

References

Dr. Md. Mashiur Rahaman
Thesis Supervisor
Professor, Dept. of NAME, BUET
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Dr. Zobair Ibn Awal
Faculty Advisor, WFSA Design Competition
Professor, Dept. of NAME, BUET
zobair@name.buet.ac.bd

Dr. Goutam Kumar Saha
Academic Advisor & Junior Year Project Supervisor
Professor, Dept. of NAME, BUET
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